Using Parameters for Interactivity

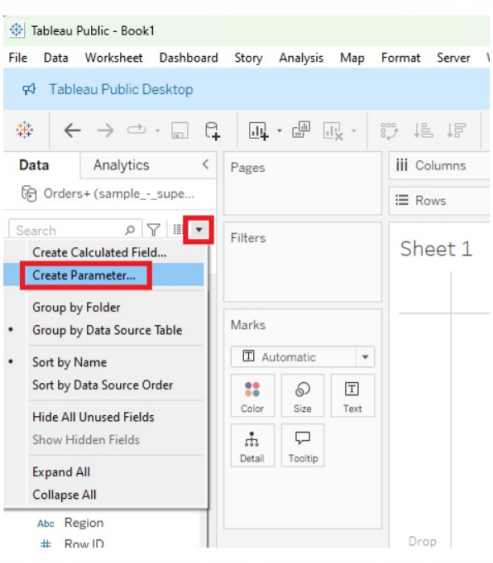
What Are Parameters?

You use a parameter as a variable to replace a constant value in your workbook visualizations. Parameters typically take the form of a string, number, or date.

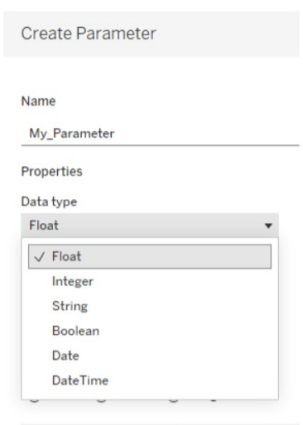
After you create a parameter, you then use it in either a calculation, filter, reference line, or reference band. Once the parameter has been applied to such an element, you can then adjust it by either using a parameter control, a parameter action, or by setting it up as a dynamic parameter that automatically updates.

Creating a Parameter

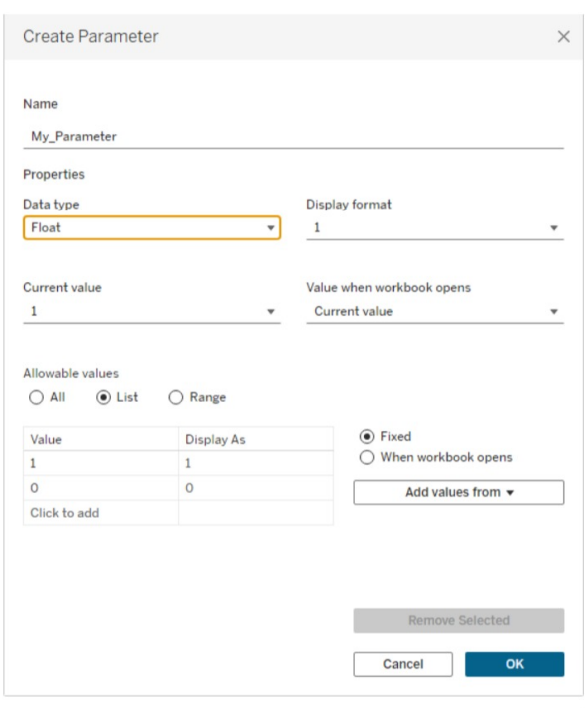
You create a parameter from the **Data** pane by clicking the little drop-down arrow and selecting **Create Parameter**.

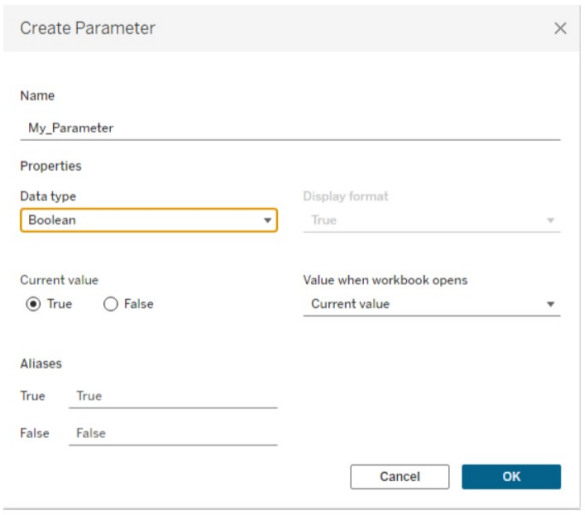


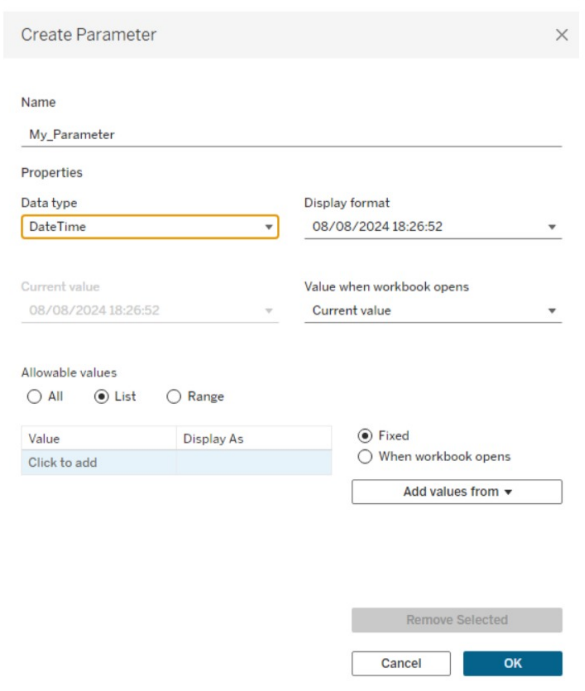
You then give your parameter a suitable name and configure its properties, such as the data type.



Other properties, such as the display format, the parameter's current values, the default value when the workbook is opened, and the allowable values, depend on which data type you initially select.

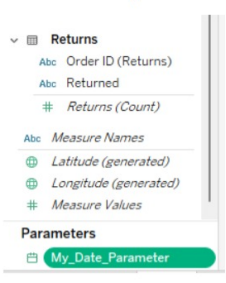






Allowable value options are: - **All**: The parameter control is a simple text field. - **List**: The parameter control provides a list of possible values to select from. - **Range**: The parameter control lets you select values within a specified range.

Parameters that you create are listed in the **Parameters** section at the base of the **Data** pane.



Using a Parameter

For a parameter to be useful, you need to associate it with an element within your visualizations, such as a calculation, a filter, a reference line, or a reference band. The parameters you create for a workbook are available globally across all the worksheets in that workbook.

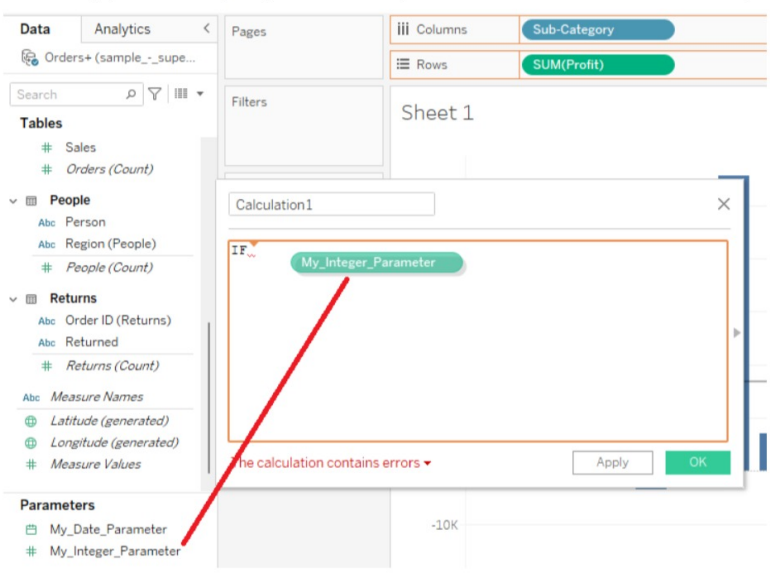
Additionally, for the parameters to be usable, they need to be configurable. You need to display a parameter control in your visualization so that users can change the values of the parameter. Alternatively, you could set up a parameter action or set up a dynamic parameter.

Using Parameters in a Calculation

Using a parameter within a calculation enables you to dynamically change the values it uses. So instead of having to manually modify each instance of a changing value, you can just change it once in the parameter control, and then all calculations that use it will be updated accordingly.

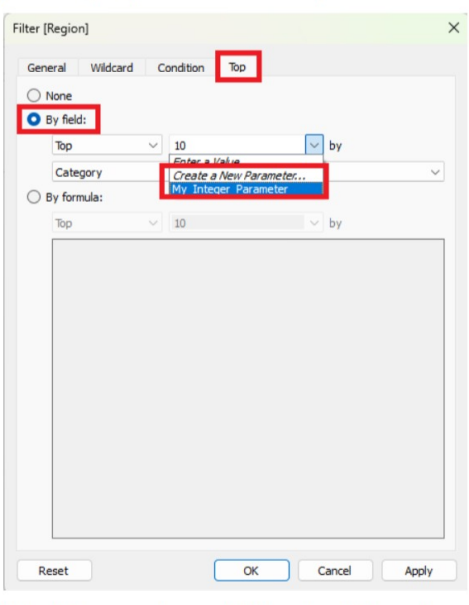
To use a parameter in a calculation, simply start entering the name of your parameter and then select it from the suggested pop-up.

Alternatively, you could drag the parameter directly from the **Parameters** section of the **Data** pane into a calculation window.



Using Parameters in a Filter

You can also use a parameter on the **Top** tab of a filter when filtering by a field to dynamically change the values of the Top *N* number instead of specifying a fixed number. To do this, just select your parameter from the drop-down list to the right of the **Top** drop-down list.



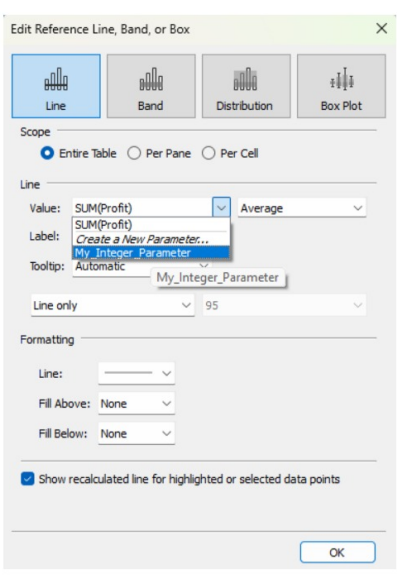
Note that you can also use this drop-down list to create a new parameter as well.

Using Parameters in a Reference Line or Reference Band

You can also use a parameter as a method to dynamically change the values in a reference line or reference band from the Analytics tab.

So, instead of displaying a reference line at a fixed point on an axis, you could reference a parameter, and then use the parameter control to modify the reference line.

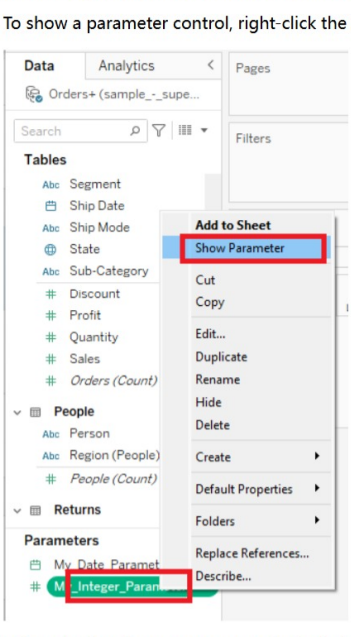
For the **Line** and **Band** options in the **Add Reference Line, Band, or Box** dialog box for either a reference line or reference band, you can select your parameter from the **Value** drop-down list, and then the line will be drawn at the value specified in the parameter control.



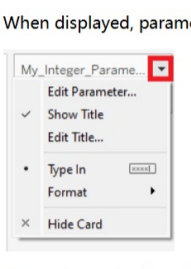
Show a Parameter Control

You use a parameter control in a visualization to enable you to modify the value of the parameter, so you need to show this control.

To show a parameter control, right-click the name of a parameter in the **Data** pane and choose **Show Parameter**.



When displayed, parameter controls look similar to a filter card and are displayed in a pane to the right of the visualization window.



Parameter controls can be displayed on worksheets and dashboards, and they are saved as part of a workbook when you save your workbooks to Tableau Public or to a Tableau Server.

Further Reading

For more information on using parameters to make your visualizations more interactive, read the *Use Parameters to Make Views More Interactive* article from Tableau Help.

Author(s)

Steve Ryan